

Reinforcement Learning Fine-Tuning of a Small Language Model for Faithful Financial Reasoning over Financial Reports

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1. Motivation

Current Problem

Large Language Models are increasingly used for

- investment analysis
- financial report understanding
- balance sheet interpretation
- ratio analysis
- earnings report summarization

However,

they frequently

- hallucinate financial values
- perform incorrect calculations
- use wrong figures from tables
- ignore numerical consistency
- produce logically inconsistent reasoning.

Why SFT is insufficient

It does not optimize

- numerical accuracy
- reasoning correctness
- equation correctness
- consistency across intermediate steps

2. Problem Statement

Existing small language models fine-tuned on financial question answering datasets often generate numerically inconsistent or logically incorrect answers despite producing fluent text. This project proposes an RL-based fine-tuning framework that optimizes reasoning correctness, numerical consistency, and factual grounding beyond conventional supervised fine-tuning.

3. Objective

Develop a reinforcement learning framework that improves

- numerical reasoning
- arithmetic accuracy
- financial reasoning
- evidence grounding

while preserving

- language fluency
- answer completeness

4. Expected Outcomes

The RL model should

- improve Exact Match
- improve F1
- improve numerical accuracy
- reduce hallucinations
- improve reasoning correctness

without increasing model size.

5. Dataset

FinQA

a large-scale dataset with 2.8k financial reports for 8k Q&A pairs to study numerical reasoning with structured and unstructured evidence.

ConvFinQA

Multi-turn conversations. Allows sequential reasoning.

TAT-QA (Tabular And Textual dataset for Question Answering)

Contains

- text
- tables
- arithmetic reasoning

6. Model (one of them)

Qwen2.5-1.5B-Instruct

SmolLM2-1.7B

Qwen2.5-0.5B

7. Reward Function

Composite Reward :

Answer correctness, Answer correctness, Equation correctness, Evidence grounding, Reasoning completeness

8. Evaluation

Exact Match

F1

Numerical Accuracy

Reasoning Accuracy

Hallucination Rate

Inference latency